Examiner only

10. *(a)* Students were discussing the reactivity of organic compounds. One said that reactivity was due to dipoles produced by differences in electronegativity.

Show that this is not correct by discussing the difference in reactivity between alkanes and alkenes. You should include reference to the bonding in both series.

[6 QER]

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(b) In the 19th century Cannizzaro discovered a reaction that involved disproportionation. Such reactions involve the same substance being both oxidised and reduced.

The equation for the Cannizzaro reaction is shown.

$$2C_6H_5CHO + H_2O \longrightarrow C_6H_5CH_2OH + C_6H_5COOH$$

(i) State why this reaction is classified as *disproportionation*. [1]

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(ii) A chemist used 9.50 g of C_6H_5CHO in the reaction above and made 3.39 g of $C_6H_5CH_2OH$.

Calculate the percentage yield that this mass of $C_6H_5CH_2OH$ represents. [3]

Percentage yield = ... %

(iii) Cannizzaro's reaction is usually carried out using aqueous sodium hydroxide, rather than with water as shown in the equation above.

Write the equation for the Cannizzaro reaction of C_6H_5CHO with aqueous sodium hydroxide. [1]

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